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Crime Spatiotemporal Prediction With Fused Objective Function in Time Delay Neural Network

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ABSTRACT In the criminology area, to detain the serial criminal, the forthcoming serial crime time, distance, and criminal's biography are essential keys. The main concern of this study is on the upcoming serial crime distance, time, and suspect biographies such as age and nationality. In conjunction with having time delays, the dynamic classifier, like Time Delay Neural Network (TDNN) utilized to perform nonlinear techniques-based predictions. The TDNN classifier system, like Back Propagation Through Time (BPTT) and Nonlinear Autoregressive with Exogenous Input (NARX) are two prominent examples. However, BPTT and NARX techniques are unable to identify the dynamic system by using single-activation functions due to producing lower accuracy. Hence, during the training phase, the direct minimization of the TDNN error can further enhance the single activation function. Thus, this work introduces an enhanced NARX (eNARX) model based on the proposed activation functions of SiRBF via fusion of two functions of the hyperbolic tangent (Tansig) and Radial Basis Function (RBF), in the same hidden layer. If a fusion of activation functions can affect the TDNN error minimization, then fusing of the Tansig and RBF functions can produce a precise prediction for crime spatiotemporal. To evaluate the proposed technique and compared it with existing NARX and BPTT, we utilized five time-series datasets, namely, Dow Jones Index, Monthly River flow in cubic meters per second, Daily temperature, and UKM-PDRM datasets namely, "Suspect & Capture" and "Crime Plotting." The analysis of the results demonstrated that the proposed eNARX produce higher accuracy in comparison to other techniques of NARX and BPTT. Consequently, the proposed technique provides more effective results for the prediction of commercial serial crime.

INDEX TERMS Back propagation algorithm, enhanced NARX, NARX, neural network, temporal pattern recognition, quantum geographic information systems.

I. INTRODUCTION

In the determination of the real-time and spatial data, a recurrent architecture has utilized for spatiotemporal prediction [1], [2]. One of the main factors to prevent serial criminals in the criminology area is forecasting the next serial crime incident. The significant aspect of crime prediction is location or scene [3]. Location is a de facto of the

criminal incidence, which is related to regulation, offender, and target [4]. The crime spot and Geographic data of the criminal event able to provide a better conception towards the crime-related environmental component [4].

There are two ways for any crime forecasting: 1) Strictly theoretical method where the prior case utilized to forecast the upcoming criminal case, and 2) Developed technique where the behavioral model of suspects is utilized to determine the self-determining variables [5]. The geographical information system (GIS) is considered a well-known

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strategy to demonstrate geographical features, such as crime spots, etc. [6]. The GIS, with the utilization of a specific algorithm, is capable of unveiling the spatial associations among geographic and communicative material [6]. Fig.1 shows the distribution of crime related to 2010 up to 2017 in KL and Selangor. The highest number of crimes are reported for unarmed robbery and vehicle theft, whilst the lowest number of crimes are for murder and snatch theft.

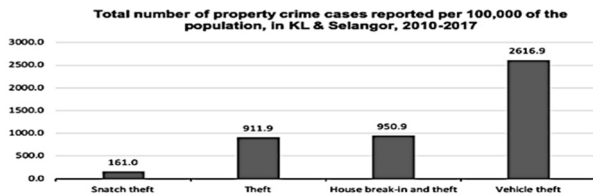


FIGURE 1. Distribution of crime cases for unarmed robbery and vehicle theft per 100,000 population in KL and Selangor.

A Neural Network (NN) is an influential data modeling approach that can indicate complicated input/output relations for identifying patterns in different data sets. The main advantage of a NN is to resolve complicated difficulties for standard technologies that not being able to be addressed with algorithmic solutions [7]. The NN has extensive applications in various research areas, and Time Delay Neural Network (TDNN) is regularly utilized to tackle the classification issue of time series for a single activation.

In general, NNs are classified into two types of static and dynamic nets. Dynamic nets conquered the issue of the non-dynamic network through dynamic recurrent nets. Dynamic nets involve two categories: 1) Network with having feed-forward associations with delays at the input layer and 2) Networks with having recurrent connections. The recurrent neural networks have the forecasting ability for upcoming values based on the previous instant values [8].

Nonstatic networks able to tackle three distinctive issues, namely Nonlinear Autoregressive with Exogenous (external) Input (NARX), Nonlinear Auto-Regressive (NAR) model, and the third type uses the two series of input and output (input-output model). The NARX technique has considered the best method among them for prediction purposes [9] due to utilizes the additional information of the prior values of the second time-series. Additionally, the Back Propagation Through Time (BPTT) algorithm is a prominent feed-forward time-delay network for time-related task difficulties [10]. However, [10] claimed that time-delay considered as the primary source for causing instability and poor performances of dynamic systems [10]. With regards to addressing time delays, the dynamic classifier, like Time Delay Neural Network (TDNN), can perform significantly in nonlinear techniques-based predictions.

The TDNN classifier system, like Back Propagation Through Time (BPTT) and Nonlinear Autoregressive with Exogenous Input (NARX), are ideal examples of these systems. However, BPTT and NARX techniques are unable to identify the dynamic system by using single-activation

functions due to producing lower accuracy. Hence, during the training phase, the direct minimization of the TDNN error can further enhance the single activation function.

A fusion mechanism of activation functions can affect the error minimization of the TDNN [7]. Thus, this mechanism is attributable to produce better precision in the crime spatiotemporal prediction task. Consequently, this study aims to introduce an enhanced NARX (eNARX) model with fused activation functions, namely hyperbolic tangent (Tansig) and Radial Basis Function (RBF) at similar hidden layer for a commercial serial crime spatiotemporal. The fusion of these functions can produce better crime prediction in terms of accuracy.

The present study is structured into five parts. State of the art is described in part II, and the materials and methodology are given in part III. The experimental results are provided in part IV, results and analysis part V and finally, the conclusions and future work are drawn in part VI.

II. RELATED WORK

A. ARTIFICIAL NEURAL NETWORK

Artificial Neural Networks (ANNs) are a distinctive way to process data that has developed continuously for about the last fifty years. The ANN solves specific problems due to having a considerable number of immensely unified processing components that act as cooperative neurons. The ANNs are applied in different fields of classification, regression, and Multi-layer perceptron's approximation [6].

In [11], the ANN has first introduced and theorized. These networks created once an ensemble of perceptron works collectively in a single system. Another ANN structure called Multi-Layer Perceptron (MLP) is widely used by modelers, which is a feed-forward neural network (FNN) [12]. Usually, a feed-forward network constructed based on three layers of input, hidden and output layers. The input layer works as the predictor variables, while the output layer acts as the criterion variables. From the input layer to the output layer of the network in a direction from left to right, all information transfers sequentially. All layers consequently delay transmitting the information of the previous layers before calculating and substituting its value. An MLP utilizes the back-error propagation for training as a supervised learning technique [13].

In different fields of science and engineering [9], Artificial Neural Networks have also efficiently applied for various issues like classification, regression, and approximation [6], [14]. Throughout the years from 1970 to 1980, perceptron-like neural networks that utilized approximation of nonlinear functions operated with an integral of multi-layer neurons and nonlinear signal processing. A significant research extension has accomplished in this specific domain, and the results included successful determinants of using neural networks for prediction [15], [16].

Another research on dynamic neural networks [17] examined NARX and evaluate it against a Back Propagation Neural Network (BPNN) to acquire an appropriate forecasting model

for the short-term wind speed with time-series attributes such as wind direction and humidity. The outcome of [17] showed that the NARX technique had higher efficacy concerning the BPNN model. Therefore, the neural network technique has shown a great understanding of real data when predicting the short-term wind speed, which can be employed to optimize the wind energy conversion system [17] additionally.

B. TIME DELAY NEURAL NETWORK

Time Delay Neural Network (TDNN) is an ANN approach in which each node connects entirely through directed connections [18]. Target tags are used at all-time steps for determining errors in supervised learning. On the other hand, errors that are signified by the overall amount of deviation in the activation will perform calculation through the output nodes network by the equivalent target tags. With the motivation to minimize network error, the system considers training algorithms extensively. For forward-feedback neural network methods, the Levenberg-Marquardt (LM) algorithm can train TDNN effectively as well [18].

One of the prominent TDNN classifier systems is NARX. The NARX model that has extensively applied in various fields can denote different nonlinear dynamic behavioral patterns [17]. NARX is equivalent to Turing machines, and its primary application is in other control systems [6]. Therefore, NARX has been a well-recognized model not only for its precision modeling but also due to its uncomplicated training.

The study of [19] showed that the weights hyperparameters have a significant impact on the training performance of the NARX model. Therefore, using the LM algorithm in the NARX model is advantageous due to the enhancement of bias and weight in the training phase. Additionally, the LM provides faster network convergence with few hundred-weights, which could address the limitation of the function approximation [19].

Figure 2 demonstrates the general structure of the NARX network that includes the individual (Tansig) at the first hidden layer. Similar types of technologies used in the LM algorithm with the withdrawal of other hidden layers than three and four for the serial PDRM-UKM crime case. The algorithm expands with the three training inputs, one testing input, and one outcome for the forecasted value of a specific incident.

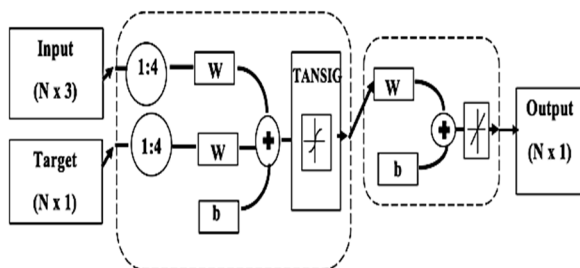


FIGURE 2. The general structure of the NARX model.

The letter of [20] examined their suggested neural network technique through training with a selection of back-

propagation algorithms, namely, the Gradient Descent (GD), stochastic Gradient Descent (SCGD), Levenberg-Marquardt (LM), and Bayesian Regularization (BR). Their findings indicated that the GD algorithm converged slowly compared to other algorithms of LM, SCGD, and BR. The LM algorithm required the smallest average time compared to BR, SCGD method. Therefore, this evidence dictated that the LM algorithm outperformed other states of the art functions with the fastest convergence time during the training performance. Predictions with the fabrication of activation functions in the neural network algorithm [20] have outlined that the LM algorithm has given the most remarkable efficacy rates. Hence, fast convergence and good predictive quality are a pair of essential indicators that make the LM as the best choice for training the neural network [20].

Another TDNN classifier system is BPTT [21]. The BPTT, a class of recurrent neural nets, utilized feedback connections intensively. Tentatively, BPTT with feedback connections embraced the time-based recounting technique that adhered to a systematic multi-layered feed-forward network [21]. In a glance, a feedback mechanism initiated better network transformation into the feed-forward system such that it inserted an additional layer at each time-step. With the introduction of a fixed quantity of layers into a feed-forward network, the fluctuation of the weight values in the recurrent network was re-assessed in a more conclusive manner [22].

In the work described in [23], two methods, those of NARX and BPTT, were utilized for the forecast of forthcoming sequential crime locations, and a comparison between these models showed that the NARX method presents higher precision when predicting the next serial crime distance compared to the BPTT model. According to the study in [23], a couple of transfer functions, namely, Tansig and RBF, were fixed in different hidden layers to boost the performance of the NARX model by using two training algorithms, namely, LM and SCGD. These transfer functions are known as bi-transfer functions, which are utilized in the LM and SCGD algorithms to predict the biography of the next suspect in the commercial serial case. The outcomes of this study present that the combination of these activation functions in different hidden layers in both training algorithms showed better performance compared to using an individual activation function.

C. SPATIOTEMPORAL DATA PREDICTION

Crimes have been part of societies since the first laws were approved. For many career options, education on crimes and criminals is a need. The research on crime is beneficial to tackle various criminal cases through investigating, fusing the complete relationships and differences among law, psychology, and social science [24].

For the prediction of near-future crimes, an analysis of a predictive model to map crimes by employing big data analysis specifies the prediction framework on the Hadoop platform [25]. Furthermore, this analysis also identifies hotspot methods [26]. The statistical implication and the identification of hotspot clusters in [26] can be applied as a basis

of a predictive algorithm for hotspot visualization. Also, through referring to backdated events, the areas of high crime-incidences are recognized. Therefore, at the identified locations, the applicable police resources can be arranged. For processing a large number of data within the half of the time regarding other systems, software [25] can join the ability of Hadoop as well [25], which have been studied earlier. Finally, the prediction that deployed an ANN method has been based on the classifications and regression-tree predictive specifications. According to the analysis [25], the mapped clusters provide sufficient data that have been analyzed. For testing the data fitness, the gamma procedure has been used. Thus, the ANN-based Hadoop model has been implemented by utilizing the obtained results from the gamma test [25].

Upon detecting various types of crime, computational criminology is beneficial that utilizes computer power, i.e., through patterns and expanding patterns, felony producers, attractors, and cybercrimes [8]. Dynamic information system analysis and computational topology can optimize the algorithm development process. Crimes do not occur arbitrarily; also, they conform to instructions similar to the non-criminal behaviors. Hence, focusing on computational criminology is vital for expanding the awareness of rules to generate the appropriate procedures for the identification of risky parts [27]. Upon dealing with inconsistency, the computational criminology research can forecast a future crime period attained from the collection of prior data, which is inter-related to serial felons. Predicting the places of the upcoming crime incidents is a crucial source of information for law execution, from both a tactical and strategic aspect [28].

The study of [29] proposed the crime-forecasting model for planning and policy decisions, which reinforces the police strategic resource distributions. Crime prediction for police forces has been made possible through GIS applications [29]. Their gamma testing application with neural network modeling includes the full original features of neural networks, Box-Jenkins ARIMA multivariate transfer models, exponential smoothing, and co-integrated econometrics. This approach measured the spatial crime clusters by utilizing a Gini-index-based method for time forecasting of the construction. It also constructed the fluctuation impact of the data aggregation level (geographic scale) based on forecasting accuracy [29].

Historical data are valuable to forecast crime incidence periodically as it can explicitly indicate the upcoming value of the location and time for the next crime. A survey paper [30] highlighted several crime prediction applications by employing effective computational techniques. They precisely utilized and reviewed several data-mining forecasting approaches to improve the precision and speed up the crime forecasting [29].

In conclusion, among the neural network categories, the dynamic networks are showing outstanding performance due to its ability to tackle the non-dynamic nets issue by dynamic recurrent nets. Time Delay Neural Network (TDNN)

usually is utilized to solve time series classification [22]. Among the TDNN classifier system, several works stated that the NARX technique produces better forecasting in comparison to other dynamic models and recurrent networks [6], [17], [19], and [23]. Furthermore, the choice of training algorithm has the main impact on the performance of the NARX technique [20]. Among the existing algorithm, the LM produces better performance due to having fast convergence during the training phase of the network. Also, this algorithm used in the NARX model is beneficial with the enhancement of bias and weight in the training stage. The geographical information system (GIS) is considered a well-known strategy to specify the exact location of the criminal case while it utilizes a specific algorithm [6].

III. MATERIAL AND METHODS

This section divides into three subsections. First, we describe the principle and proposal framework of predictions that use an eNARX classifier. Next, an intensive discussion about a description of the datasets and its attribute, the pseudocode, and the pre-processing methods handled in the second section. After explaining our proposed model in the third section, a series of empirical, experimental results derived from before and after the proposal of eNARX model has been inferred and synthesized in the final section.

A. THE PRINCIPLE AND PROPOSED FRAMEWORK FOR TEMPORAL PREDICTION

Figure 3 shows the framework with four key stages that have studied, namely: initial, pre-processing, classification, and prediction. At the initial stage, the time series data have gathered from multiple sources to conduct the research.

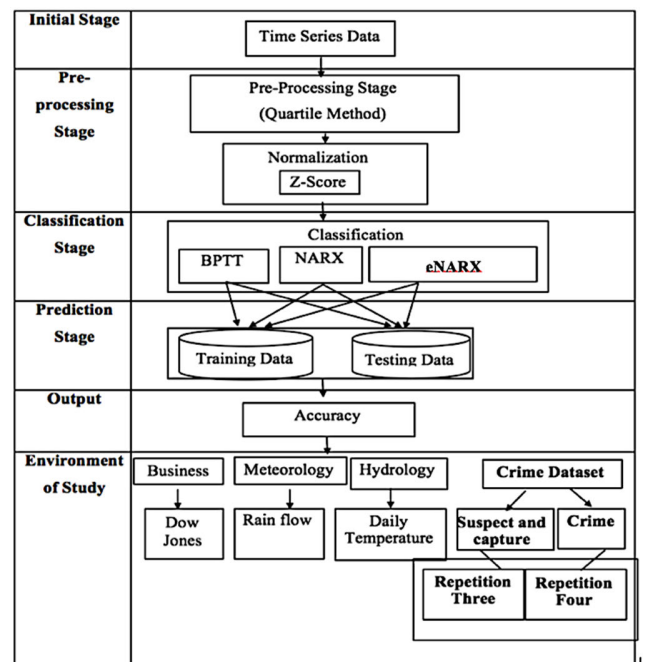


FIGURE 3. The principle and proposed framework for prediction.

For removing outliers from datasets, the Quartile method has chosen and applied. A popular normalization method, namely Z-Score and max filtering, has been used to rescale and replace missing values in the crime time series data comprising the distance-attributes during a pre-processing phase. In the classification stage, these pre-processed data utilized BPTT, NARX, and eNARX to forecast the future values of serial crime events in the crime dataset and time series benchmarking datasets. Later, based on results found from the relative study, the NARX model has been enhanced by fusing it with RBF and Tansig to produce SiRBF. These eNARX models have been analyzed and evaluated to obtain the best results of prediction by using UCI benchmark datasets with Dow Jones Index, Monthly River Flow and Daily Temperature, and one self-collected dataset (Malaysian crime spatial). Finally, in the prediction stage, a prediction has been performed by splitting data between training and a testing set.

B. DATASET DESCRIPTION

In this study, Malaysian crime spatial datasets and three benchmark time series were used. The two crime spatial datasets, namely, (1) Suspect & Capture (SCC) and (2) Crime Plotting (CP), utilizing in the SPBS system Neo (2014), were combined for predicting the next time, location and suspect of a serial crime with specific attention. Ethical approval received from the Royal Malaysia Police (RMP) in 2013, covers Wilayah Persekutuan and Selangor states for mapping through the CP datasets. The descriptions of two crime spatial datasets and three-time series benchmark datasets are shown in Tables 1, respectively.

TABLE 1. RPM-SPBS crime spatial datasets descriptions.

Name of Datasets	Suspect & Capture	Crime Plotting
Authorship	RPM	SPBS
Dataset Characteristics	Multivariate	Multivariate
Attribute Characteristics	Categorical, Integer	Categorical, Integer
Instances value	105934	136407
Attributes number	14	14
Missing values	Yes	Yes

The overall instance value of the “Suspect & Capture” and “Crime Plotting” datasets are 105,934 and 136,407, respectively. The datasets mentioned above include 14 attributes. Examples of its attributes are age and nationality of criminals, distance, and time of crime incident.

Before the prediction stage, the pre-processing step conducted in advance before predicting the next crime time series neural network. Preparing the data for the next serial-crime distance prediction, it selected Name, ID Number, and Case Number attributes from the “Suspect & Capture” dataset. It carefully chose the Latitude & Longitude and the number of cases from the “Crime Plotting” dataset to determine the distance value for every serial case.

The quantity of cases in these datasets works as a primary step for merging the latitude and longitude characteristic from the “Crime Plotting” to the “Suspect & Capture” dataset. As this study based on serial crimes, each serial case is indicated based on the number of repetitions. We assume the number of repetitions for each criminal case obtained from the duplications of the name and identification number of the suspect. Therefore, we only select the number of repetitions from three and above to be trained in the eNarx model. After combining the two datasets mentioned above, the instance value of the combined dataset for a repetition of three and above, based on the number of duplicates, become 10415.

In this study, the explanation of the pre-processing stage and also the merging data steps are described as follows:

Algorithm 1 Integrating Datasets Steps

Begin

Step 1: Open “Suspect & Capture” and “Crime Plotting” datasets with an external data option.

Step 2: Fusing the two datasets mentioned above through utilizing query design.

Step 3: To obtain duplicate numbers by query wizard.

Step 4: Retain the duplicates with occurrences of three or above.

Step5: Calculate distance between the serial incidents of X(Latitude(a), Longitude(b))-Y (Latitude(a), Longitude(b)) by:

$$Distance(K, T) = \sum_{K,T}^{n,m} \cos^{-1}((\sin(\pi - \theta^{Lat(a)}) \times \sin(\pi - \theta^{Lat(b)}) + \cos(\pi - \theta^{Lat(a)}) \times \cos(\pi - \theta^{Lat(b)}) \times \cos(\theta^{Lon(b)} - \theta^{Lon(a)})) \times R[31],$$

where K= the value of the record of SCCP criminal between [a, n], n = the records numbers, T = the frequencies numbers among [a, m], m = the repetitions value, Lat and Lon are the latitude and longitude of two serial criminal events, and R = 6371 Km (the Earth’s radius).

End

C. THE PROPOSED ENHANCED NARX MODEL

In this research, the dynamic neural network NARX model was used to forecast the time-series above datasets, which comprise the benchmark datasets and our self-collected SCCP datasets. Then, to enhance the NARX technique for better precision, it was merged by the two activation-functions as Tansig and RBF in the same hidden layer. This step was accomplished for testing and improving the performance of the NARX model by using the proposed SiRBF in NARX. Fig. 4 shows the flowchart for the eNARX prediction model for the time-series datasets.

D. PRE-PROCESSING STEPS

We performed the pre-processing phase by using the max filter for noise removal. We also used Z-score normalization for avoiding attributes with big scales of data from outweighing

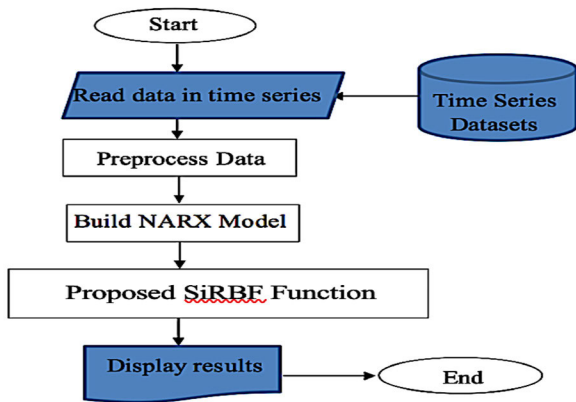


FIGURE 4. The flowchart for forecasting of the eNARX serial crime events and the time series datasets based on the proposed SiRBF function in the NARX classifier.

attributes with smaller scales of data for the distance-based methods [32]–[34] and the Quartile filter for outlier filtering.

1) MAX FILTERING

In this research, three and four repetitions of the serial occurrence have selected from more than thirty-three repetitions. The Max filter has sorted every null value, i.e., the serial crime distance value. According to the vectors X noise removal, such a step is crucial; once the null values are discarded, it substitutes the maximum value in place of the null value of the remaining elements in each the row vector of column maxima for a particular X matrix [33].

2) NORMALIZATION

The normalization scheme utilized to eliminate the data redundancy as well as organizing the data efficiently to make the time series data readable for the classifiers. Undertaking several essential pre-processing steps can optimize the training inputs and targets of the neural network efficiently. Usually, for neural networks or measurements of distance, normalization is the appropriate method. For example, the nearest neighbor, agglomerative, and clustering method. By normalizing the input attribute, the learning process can accelerate while using a neural network algorithm for classification. Also, normalization facilitates to avoid attributes underlying big scales of data from outrival attributes with very small scales of data for distance-based methods [31]. Data normalization methods apply the z-score method, decimal scaling method, and min-max technique consecutively. In this study, the Z-score normalization technique has used the min-max normalization or the real minima and maxima values of the time series, to address outliers influence unidentified [33]. Given that the D'_c is the normalized value of the attribute D_c which is calculated through the mean, M and standard deviation, $Stdv_D$ of the D_c attribute, Eqn. (1) displays the Z-score normalization calculation:

$$D'_c = \frac{D_c - M}{Stdv_D} \quad (1)$$

The Z-score technique is appropriate once the real minima and maxima value of D_c is unknown or unrevealed. A sample of the normalization for repetition four datasets with distance attributes is shown in Table 2.

TABLE 2. Normalization sample for repetition four dataset with distance attribute.

Raw Data				Normalized by Z-Score			
Age	Nation ality	Time	Distance	Age	Nation ality	Time	Distance
36	11	23:30	0.8386766	0.4308461	-	1.4682445	-
36	11	0:01	0.3632087	0.4308461	-	-	-
36	11	19:30	1.2304784	0.4308461	-	0.881597	-
36	11	0:30	13.399932	0.4308461	-	-	1.4646127
36	11	20:30	9.7650422	0.4308461	-	1.0282589	0.7743273
36	11	15:54	1.6217838	0.4308461	-	0.3301482	-
36	11	22:00	13.399932	0.4308461	-	1.2775841	1.4646127
36	11	21:00	13.399932	0.4308461	-	1.1309222	1.4646127
36	11	12:36	5.9396462	0.4308461	-	-	0.0478638
36	11	17:40	5.9147348	0.4308461	-	0.6029394	0.043133
36	11	11:49	0.0503272	0.4308461	-	-	-
36	11	18:31	13.399932	0.4308461	-	0.2638325	1.0705499
36	11	19:40	5.0132232	0.4308461	-	0.7364017	1.4646127
36	11	20:04	0.5432241	0.4308461	-	0.8962631	-
36	11	20:04	0.5432241	0.4308461	-	0.9901268	0.128069
37	11	3:00	0.6408146	0.5607569	-	-	-
36	11	3:30	13.399932	0.4308461	-	1.508992	0.9584131
36.063	11	0.598	6.182	0.438	-1.378	-	1.4646127
0.25	0	0.339	5.672	0.032	4.586E-16	1.193	1.077

3) QUARTILE METHOD

It is noteworthy that the outlier of data mining causing data deviates or strays far away from other data of a dataset. The detection of the outliers can benefit in the quest of finding the understanding of a different area named as the criminal behavior detection. In the case of weather prediction or other

exploration in scientific databases, detecting outliers makes it essential. This research employed the Quartile method to detect outliers of datasets in three steps mentioned below:

Algorithm 2 Quartile Steps

Begin

Step 1: Compute the upper quartile: Q3-75% of the data in the data set is lower than this.

Step 2: Compute the upper quartile: Q3-25% of the data in the data set is lower than this.

Step 3: Estimate the gap between the quartiles: $H = Q3 - Q1$:

1. If a value $< Q1 - 1.5$ && $H > Q3 + 1.5$ then $H =$ Mild outlier.
2. If a value $< Q1 - 3$ && $H > Q3 + 3$ then $H =$ extreme outlier.

End

E. NONLINEAR AUTOREGRESSIVE WITH EXOGENOUS (EXTERNAL) INPUT (NARX) MODEL

This study combines three multilayer feed-forward, Recurrent, and time-delay neural networks to create a NARX network in the time-series prediction [23], [35]. In the area of criminology, the forecast for the upcoming serial crime time is a significant aspect. Here, the NARX model works effectively for predicting the time successfully.

In our research, a feed-forward network of two-layers has employed for a NARX network having a hidden-layer that uses the Tansig transfer function. Besides, the output layer contained a linear transfer function. Tapped delay lines (d) used to store the prior input $x(t)$ and output $y(t)$ value-sequences in the network. The data-training has loaded into the network. Next, tapped delay lines that contained four tapped delays that utilized for the input and output. The $x(t)$ and $y(t)$ sequences are the two inputs in the series-parallel network. Of which, the sequence $y(t)$ is the feedback signal; $y(t)$ denotes both the input and output. Eqn. (2) expresses the technique as follow:

$$y(t + d) = F(y(t), \dots, y(t - d), x(t), \dots, (t - d)) \quad (2)$$

where $y(t)$ indicates the output and also the feedback to the NARX network input, and (d) represents the tapped delay-lines that collect the $x(t)$ and $y(t)$ sequences with their prior values. Upon similar motivation from [23], [35], they claimed that the LM algorithm was more efficient in the NARX networks concerning the rest of recurrent architectures. We represent the rules of the Gauss-Newton algorithm in Eqn. (3).

$$W_{K+1} = W_K - (J_K^T J_K)^{-1} J_K e_K \quad (3)$$

The LM algorithm represents an additional estimation to the Hessian matrix to be certain that the estimated Hessian matrix $J^T J$ is invertible, as indicated by Eqn. (4).

$$H \approx J^T J + \mu I \quad (4)$$

The μ is a constantly positive combination coefficient, and I is an identity matrix. If components of the estimated Hessian Matrix Eqn. (3) are positive, then matrix H become constantly invertible. The updated rules of the LM algorithm can represent by combining Eqn. (3) and (4) to form Eqn. (5).

$$W_{K+1} = W_K - (J_K^T J_K + \mu I)^{-1} J_K e_K \quad (5)$$

Gauss-Newton and the steepest descent algorithm combination can trigger the LM algorithm to switch in the middle of these algorithms during the training procedure. If μ has values that are close to zero or very small, Eqn. (5) will approach Eqn. (3) where the algorithm of Gauss-Newton is applied. As an alternative, if μ has values that are above zero, the steepest descent method is applied by approximating Eqn. (4) with Eqn. (5).

Fig.5 demonstrates our suggested technique. The overall structure of the eNARX network contains a SiRBF transfer function that uses a fusion mechanism (Tansig with RBF) at the first hidden layer with 65 nodes, and the LM algorithms. This eNARX model uses three and four serial PDRM-UKM crime cases (3 or 4 serial means a suspect was detained three or four times at different crime scenes or events). Here three training inputs, one testing input, and one output of the expected value of the specific event are present. Fig. 6 presents the NARX model with the proposed SiRBF.

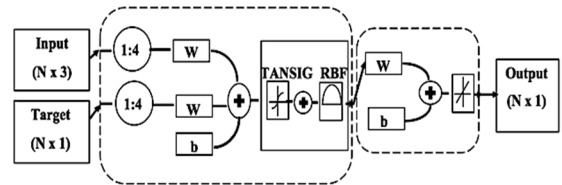


FIGURE 5. The overall structure of the eNARX model with SiRBF.

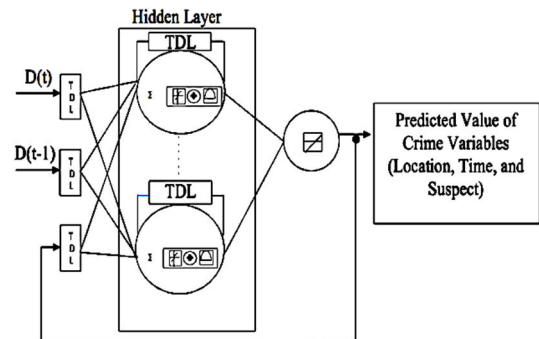


FIGURE 6. NARX model with the SiRBF objective function.

1) ACTIVATION FUNCTIONS

Functionally, RBF is a standard neural network architecture. In the LM algorithm, Tansig is the transfer function that included in both models of the NARX and BPTT. Since

Tansig function outperformed than others, it has applied to obtain initial results. Here in the study, the most suitable and accurate model has been enhanced by fusing the Tansig and RBF activation functions through using Mercer's theorem. We formulate the derivative of Tansig, RBF, and reformulate the proposed SiRBF in Eqn. (9), Eqn. (11), and Eqn. (19), respectively. The Logistic Sigmoid function can have a different range of values, which is suitable for a particular problem, as shown in Eqn. (6):

$$\mathcal{F}(x) = \frac{1}{1 + e^{-(\sigma x)}} \quad (6)$$

The most usual range of the Logistic Sigmoid function is $[-1, 1]$, which is known as the Bipolar Sigmoid function. Eqn. (7) shows the Bipolar Sigmoid function:

$$g(x) = 2\mathcal{F}(x) = \frac{2}{1 + e^{-(\sigma x)}} - 1 = \frac{1 - e^{(-\sigma x)}}{1 + e^{(-\sigma x)}} \quad (7)$$

The Bipolar Sigmoid function is intimately interrelated to the Tansig function, and here, we demonstrate the Similarity between these functions, where $\sigma = 1$, and Eqn. (8) and (9) shows the simplification of the Bipolar Sigmoid and Tansig functions, respectively.

$$g(x) = \frac{1 - e^{(-\sigma x)}}{1 + e^{(-\sigma x)}} \quad (8)$$

$$h(x) = \frac{e^x - e^{(-\sigma x)}}{e^x + e^{(-\sigma x)}} = \frac{1 - e^{(-2x)}}{1 + e^{(-2x)}} \quad (9)$$

The derivative of Tansig and RBF is defined subsequently as below:

$$h'(x) = [1 + h(x)][1 - h(x)] \quad (10)$$

$$I(x) = \sum_{K=1}^N C_K \phi(\|X - X_K\|_2), \quad X \in \mathbb{R}^S \quad (11)$$

where X_k represent the N scattered points, and C_K are the coefficients of the function after solving the linear system (12) and (13) which is states for the RBF function, as shown at the bottom of the page.

The coefficients C_K are independent of X . Thus, the partial derivatives of F can derive as in Eqn. (14):

$$\frac{\partial}{\partial X_J} I(x) = \sum_{K=1}^N C_K \frac{\partial}{\partial X_J} \phi(\|X - X_K\|_2) \quad (14)$$

By the chain rule,

$$\frac{\partial}{\partial X_J} \phi(\|X - X_K\|_2) = \frac{\langle e_j, X - X_K \rangle}{\|X - X_K\|_2} \phi'(\|X - X_K\|_2) \quad (15)$$

where the e_j is the J^{th} standard basis vector. In other words, $\langle e_j, X - X_K \rangle$ is simply the J^{th} component of $X - X_K$. By simplifying $\phi(x) = \psi(X^2)$, the argument of ψ is the squared norm, which is more pleasurable to differentiate, and the result shows in Eqn. (16).

$$\frac{\partial}{\partial X_J} I(x) = 2 \sum_{K=1}^N C_K \phi \langle e_j, X - X_K \rangle \psi'(\|X - X_K\|_2) \quad (16)$$

2) THE PROPOSED SIRBF

A transfer function determines the net weight while the activation function calculates threshold values, i.e., when the network gives the output. For this research, the proposed activation function (SiRBF) uses in the same hidden layer to form the transfer function, and the activation function is purlin, as per the rule of Mercer theorem expressed in Eqn. (17). Table 3 displays the computation of the activation functions in the eNARX technique and the effect of the improved activation functions (SiRBF) as well as outputs of the eNARX network according to the Lavenberg-Marquardt algorithm. The alteration between the two activation functions and the SiRBF is shown in Fig.7.

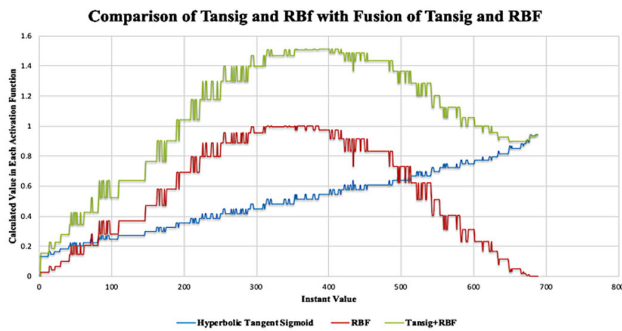
$$\begin{aligned} F_{\text{SiRBF}}(X) &= h(x) + I(x) \\ &= \left(\frac{2}{1 + e^{-2x}} - 1 \right) + e^{-(x)^2} \\ &= \frac{1 - e^{(-2x)}}{1 + e^{(-2x)}} + e^{-(x)^2} \\ &= \frac{1 - e^{(-2x)} + e^{-x^2}(1 + e^{-2x})}{1 + e^{-2x}} \\ &= \frac{1 - e^{-2x} + e^{-x^2} + e^{-x^2-2x}}{1 + e^{-2x}} \end{aligned} \quad (17)$$

$$\begin{bmatrix} I(X_1) \\ I(X_2) \\ \vdots \\ I(X_N) \end{bmatrix} = \begin{bmatrix} \phi\left(\frac{1}{\|X_1\|} - \frac{1}{\|X_1\|}\right) & \phi\left(\frac{1}{\|X_1\|} - \frac{1}{\|X_2\|}\right) & \cdots & \phi\left(\frac{1}{\|X_1\|} - \frac{1}{\|X_N\|}\right) \\ \phi\left(\frac{1}{\|X_2\|} - \frac{1}{\|X_1\|}\right) & \phi\left(\frac{1}{\|X_2\|} - \frac{1}{\|X_2\|}\right) & \cdots & \phi\left(\frac{1}{\|X_2\|} - \frac{1}{\|X_N\|}\right) \\ \vdots & \vdots & \ddots & \vdots \\ \phi\left(\frac{1}{\|X_N\|} - \frac{1}{\|X_1\|}\right) & \phi\left(\frac{1}{\|X_N\|} - \frac{1}{\|X_2\|}\right) & \cdots & \phi\left(\frac{1}{\|X_N\|} - \frac{1}{\|X_N\|}\right) \end{bmatrix} \quad (12)$$

$$I(x) = e^{-(x)^2} \quad (13)$$

TABLE 3. Calculation of the objective functions in an eNARX model.

Z-score (Value-Mean)/ Stand Deviation	Transfer Function Calculation			Outputs of eNARX Network- Based on IM
Age	Hyperbolic Tangent Sigmoid (Eqn. 9)	RBF (Eqn. 13)	Tansig+RBF (Eqn. 17)	Outputs (Eqn. 2 & 5)
-1.908	0.129	0.026	0.156	-0.188
-1.908	0.129	0.026	0.156	0.307
-1.908	0.129	0.026	0.156	-0.064
-1.908	0.129	0.026	0.156	1.883


FIGURE 7. The difference between the two activation functions of Tansig and RBF with the SiRBF.

IV. EXPERIMENTAL RESULTS

Justifying the application of our proposed SiRBF function for enhancement of NARX model, this study prepares two types of experiments. The experimental result I indicate the results for the Malaysian crime spatial datasets (repetitions three * and four *). The second experiment conducted on three benchmark datasets for namely, Dow Jones Index, Monthly River Flow (in cubic meters per second or cms), and daily temperature (in °C). Those datasets are involved in various problem disciplines such as business, hydrology, and meteorology, respectively.

A. EVALUATION METRICS

Different statistical metrics are available to authenticate the implementation of the models. However, the most frequently employed benchmark metrics of the regression coefficient (R), Mean Square Error (MSE), and Root Mean Square Error (RMSE) are applied in this study. R criterion has shown in Eqn. (18).

$$R = \frac{R_{PA}}{\text{std}(P) \text{std}(A)}, \quad (18)$$

where P and A are forecasted and are the real targets of the serial criminal and three benchmark datasets respectively, R_{PA} = the covariance between P and A , $\text{std}(P)$, and $\text{std}(A)$ represent the standard deviation of P and A , correspondingly.

The regression coefficient is fluctuating between 0 and 1. A value closes to “0” corresponds to a deviating output. Otherwise, as it approaches, “1” refers to an adaptable output that adapts to the goal.

The MSE is an extensively utilized criterion that is applied to obtain the alteration among values acquired by the specified model and the anticipated real values. A smaller MSE value implies that the method is further precise and trustworthy. The MSE standard is shown in Eqn. (19).

$$MSE = \frac{1}{N} \sum_{i=1}^N (A_i - P_i)^2, \quad (19)$$

where N is the number of samples predicted, A and P denote values of the actual and forecasted respectively. For constructing the Root Mean Square Error (RMSE), the residuals are defined in the first step. Residuals calculate the dissimilarity between real values and the RMSE, which can be computed using Eqn. (20).

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (A_i - P_i)^2}{N}} \quad (20)$$

The forecasted value can be either positive or negative that denotes it below or overestimation from real value. The RMSE can be derived as follows: first is square the residuals, second is average the squares, and third is obtain its square root.

B. EXPERIMENTAL RESULTS I

In the initial step of this research, a dynamic neural network was applied for the time series NARX and BPTT approaches. The classification phase is employing the multilayer perceptron, including backpropagation (MLBP), which was checked through MATLAB to forecast the next serial-crime cases.

According to the objective of this research, a single activation forecast for the next serial-crime has taken into consideration. The goal of a single activation is to calculate each attribute of time, distance, nationality, and age from the “Suspect & Capture” and “crime plotting” datasets; by holding three attributes as the training data and one character as the goal.

As the trial data for recurrences three and four contained outliers, the Quartile method has used to eliminate outliers in every repetition. The earliest amount of instant values for repetitions three and four have been found as 3375 and 800 respectively, and after seizing the zero value and missing values resulted from the Quartile method, the number of repetitions for the three and four datasets was reduced to 1668 and 692 respectively.

After deducting outliers from the two “Suspect & Capture” and “Crime Plotting” datasets of repetitions three and four, those acquired and filtered data were then trained on the three models called BPTT, NARX and eNARX subsequently. Finally, out of four attributes of distance, age, nationality, and time, each experiment designated only one attribute as the target value at a time in both the recurrence of three and four datasets.

1) EXPERIMENTAL SETUP

For this research, various amounts of hidden layers, from 10 up to 100, have been assessed for repetitions three and four, to achieve the most appropriate number of hidden neurons. This assessment shows that several hidden neurons between 65 to 70 represented the most exceptional performance in both repeats. Thus, hidden layers were set to 65 to 70 by a delay of four, and data of each repetition were allocated into three segments for training, validation, and testing at 70%, 15% and 15% respectively [36].

2) EXPERIMENTAL AND ANALYSIS FOR CRIME SPATIAL DATASETS

To evaluate the execution of three models of proposed eNARX, NARX, BPTT, crime spatial datasets were applied in the training, validation, and testing stages of these models. An evaluation of the accuracies for repetitions three and four with their means and standard deviations represented in Table 4. Fig.8 displays an evaluation of the precision values for the BPTT, NARX, eNARX models for repetitions.

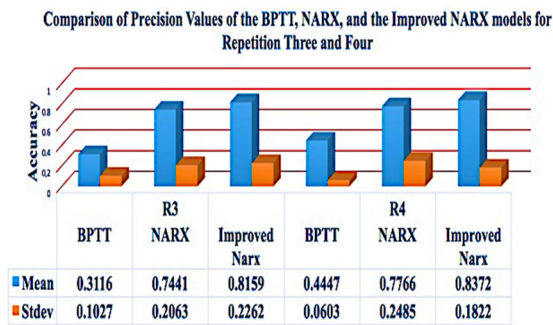


FIGURE 8. Comparison of the precision of the BPTT, NARX, and eNARX models for repetitions three and four.

TABLE 4. Descriptions evaluation of performance precision for repetition three and four with distance, age, nationality, and time target attribute for BPTT, NARX, And eNARX models.

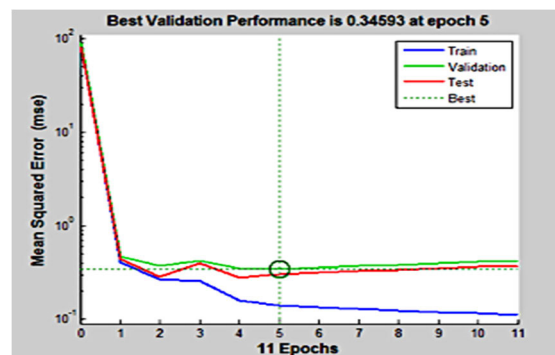
Dataset		BPTT Model				NARX Model				eNARX Model			
Repetition Three	Attributes	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²
	Distance	0.1863	0.9800	0.9900	0.0347	0.8729	0.2426	0.4925	0.762	0.8966	0.1966	0.4434	0.8039
	Age	0.4003	0.8567	0.9256	0.1602	0.8015	0.3677	0.6064	0.6424	0.8408	0.2944	0.5426	0.7044
	Nationality	0.3907	0.8715	0.9335	0.1526	0.8636	0.2586	0.5085	0.7458	0.8899	0.2089	0.4571	0.7919
	Time	0.2692	0.9290	0.9639	0.0725	0.4383	0.8430	0.9181	0.1921	0.4850	0.7661	0.8753	0.2352
	Mean	0.3116				0.7441				0.8159			
	Stdev	0.1027				0.2063				0.2262			
	Attributes	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²
	Distance	0.3837	0.8631	0.9291	0.1473	0.8303	0.3166	0.5627	0.6894	0.8849	0.2335	0.4833	0.7830
	Age	0.4549	0.8115	0.9008	0.2069	0.9962	0.0079	0.089	0.9924	0.9970	0.0058	0.0763	0.9940
Repetition Four	Nationality	0.5239	0.7428	0.8619	0.2745	0.8599	0.2792	0.5284	0.7394	0.8919	0.2044	0.4521	0.7955
	Time	0.4163	0.8469	0.9203	0.1733	0.4199	0.9572	0.9784	0.1763	0.5749	0.7049	0.8396	0.3305
	Mean	0.4447				0.7766				0.8372			
	Stdev	0.0603				0.2485				0.1822			
	Attributes	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²

The experimental result I indicate that the mean of precision of proposed eNARX, NARX, and BPTT are %83.72, %77.66, and %44.47, respectively. Therefore, the proposed eNARX outperforms the other two techniques in terms of higher accuracy for the prediction of the next serial crime. Consequently, as eNARX model produces higher accuracy, so we select the best accuracy for each of four utilized attributes of distance, age, nationality, and time. The best eNARX performance for each distance, age, nationality, and time attributes were in hidden layers 70, 69, 69, and 66, respectively. Fig. 9 demonstrates the performance plotting of repetition three with the targets of (a) distance, (b) age, (c) nationality, and (d) time vs. the number of iterations (Epochs). The training of the network for each target of distance, age, nationality, and time stopped at the iterations of 11, 12, 11, and 13 after the validation error increased steadily from 6th iteration, respectively.

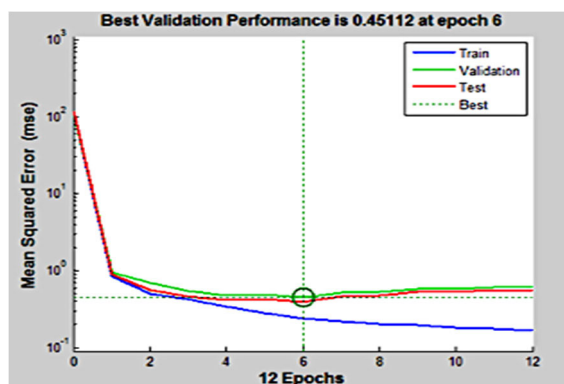
C. QUANTUM GEOGRAPHICAL INFORMATION SYSTEM (QGIS)

In the current research, google earth was employed to illustrate the actual and predicted values. Since the outcome of this study showed that eNARX gives a higher accuracy compared to BPTT, and NARX models, the anticipated value from eNARX model has been utilized for displaying the interval amid the real and forecasted values of repetitions three and four [30], [37]–[40].

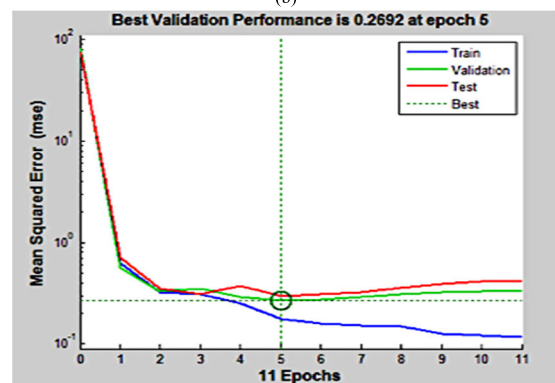
Among every repetition three and four, only one case randomly chose. Each of these repetitions has included three and four distance values. Here, distance values for the first two incidents of repetition three along with the first three incidents of repetition four have computed by finding the



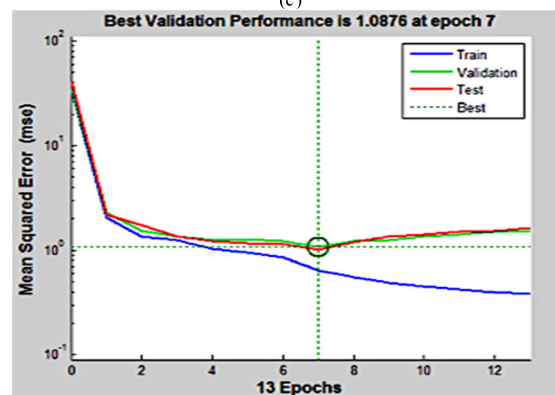
(a)



(b)

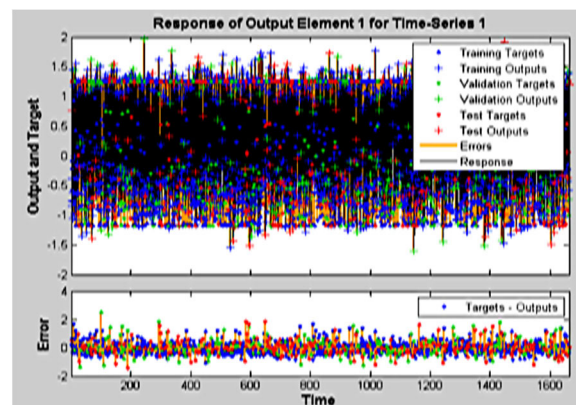


(c)

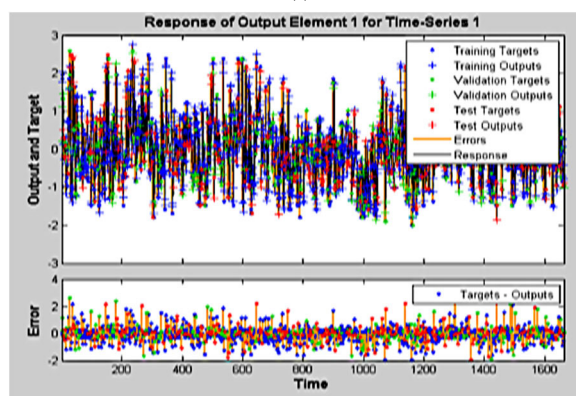


(d)

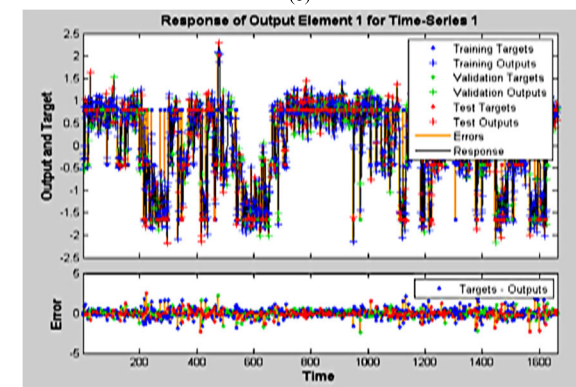
FIGURE 9. Performance plotting of repetition three with the targets of distance (a), age (b), nationality (c), and time vs. the number of iterations (Epochs).



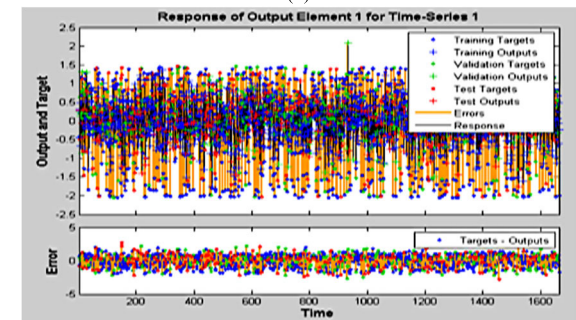
(a)



(b)



(c)



(d)

FIGURE 10. Displays LM (SiRBF) with the target of distance, (b) LM (SiRBF) with the target of age, (c) LM (SiRBF) with the target of nationality, and (d) LM (SiRBF) with the target of time. Moreover, the time points that were chosen for training, testing and validation are shown.

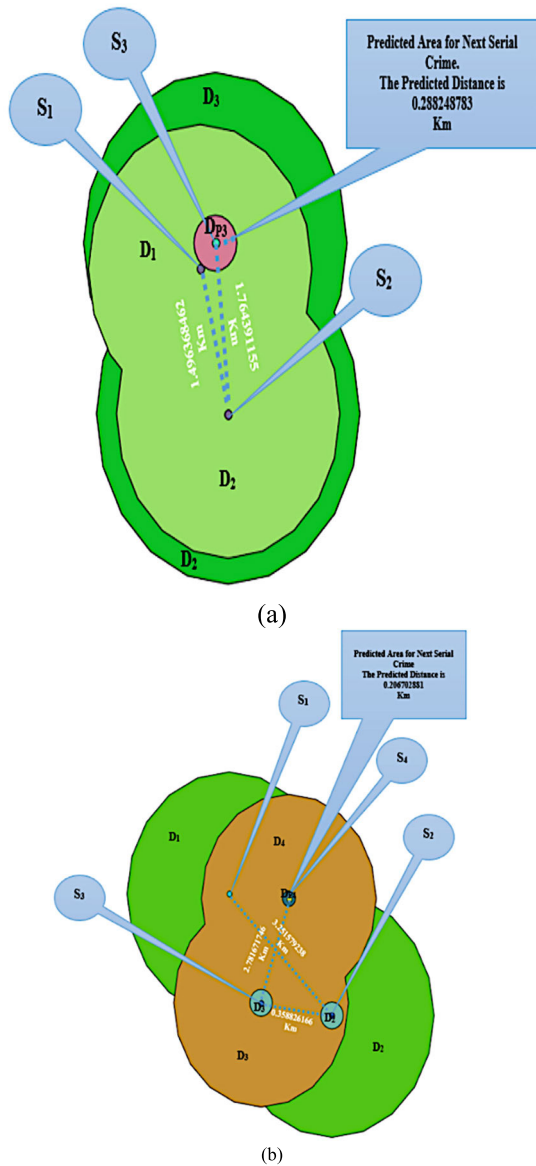
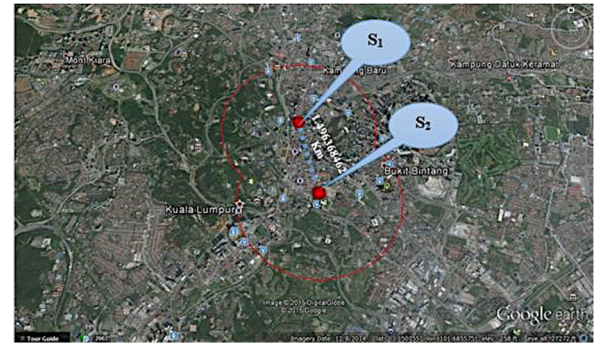


FIGURE 11. Distinctions between the predicted distance value with the first two distance values for (a) repetition of three and (b) repetition of four serial criminal incidents for the areas of Selangor and Wilayah Persekutuan by QGIS.

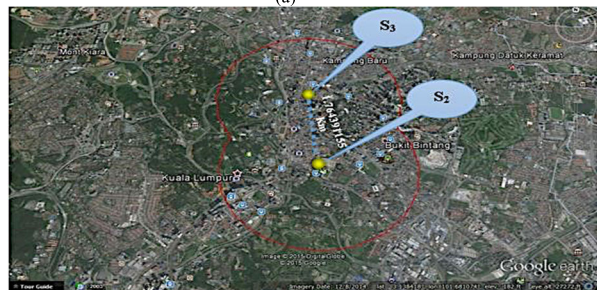
distance between the serial incidents of two serial criminal events subsequently by the equalization mentioned in [28]. Next, the distance values for the third and fourth incidents were predicted using the eNARX model.

The QGIS was utilized to generate the buffering throughout the longitude and latitude of each serial event for case repetitions three and four based on their calculated distance values. We observe that the buffer illustrates the gaps amongst the earliest two distance values (D_1 , D_2) with the forecasted distance value (DP_3) in case of repetition three and the most initial three distance values (D_1 , D_2 , D_3) plus its predicted distance value (DP_4) in case of recurrence four.

Besides, to avoid the upcoming serial incident, Google earth has been utilized to display the predetermined zone



First distance (D_1) between S_1 and S_2 of the serial incidents for the repetition of three at the focus of 27272 feet



Second distance (D_2) between S_2 and S_3 of the serial incidents for the repetition of three at the focus of 27272 feet



Third distance (D_3) between S_3 and Sp_4 of the serial incidents for the repetition of three at the focus of 27272 feet

FIGURE 12. Distinctions between the predicted distance value with the first two distance values for the repetition of three serial criminal incidents at Selangor and Wilayah Persekutuan, where (a) the red circles that buffer around both serial one and two shows the calculated distance between these points, where each point occurred at night, and the distance value is 1.496368462 km, (b) the red circles that buffer around both serial two and three show the calculated distances between these points, where each point occurs in the morning, and the distance value is 1.764391155 km, and (c) the green circle that buffers around serial three shows the predicted distance value by enhanced NARX. The predicted distance value from serial three is 0.288248783 km.

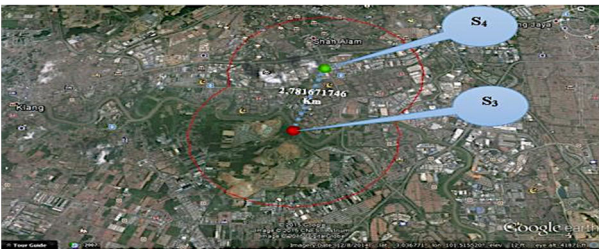
of every serial incident and the gaps between the real and forecasted distance. On the Google Earth map, the blue color dotted points of repetitions three and four identify the serial crime time event throughout the day. We denote yellow, green, and red color points to indicate the time of day, morning, afternoon, and night respectively. Fig. 11 demonstrates buffering around each of the points of repetitions three and four, to observe the gaps in the distances among the real and forecasted values using QGIS. Fig. 12 and Fig. 13 illustrate the gaps between the predicted distance value with the first two distance values for (a) repetition of three



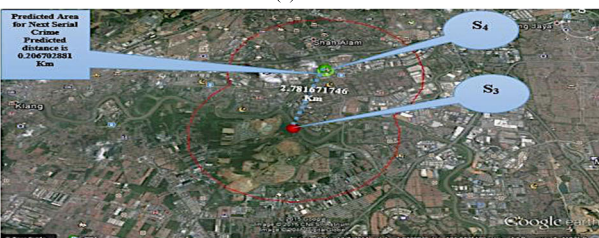
First distance (D_1) between S_1 and S_2 of the serial incidents for the repetition of four at the focus of 41871 feet
(a)



Second distance (D_2) between S_2 and S_3 of the serial incidents for the repetition of four at the focus of 41871 feet
(b)



Third distance (D_3) between S_3 and S_4 of the serial incidents for the repetition of four at the focus of 41871 feet
(c)



Third distance (D_4) between S_3 and S_4 of the serial incidents for the repetition of four at the focus of 41871 feet
(d)

FIGURE 13. Distinctions between the first two distance values with the predicted distance value for the repetition of four serial criminal events at Selangor and Wilayah Persekutuan (a) the red circles that buffer around both serial one and two show the calculated distances between these points, where each point occurs in the afternoon. The distance value is 3.251579238 km, (b) the red circles that buffered around both serial two and three show the calculated distance between these points, where points two and three occurred in the afternoon and night, respectively. The distance value is 3.251579238 km, (c) the red circles that buffered around both serial two and three show the calculated distances between these points, where points three and four occurred at night and afternoon, respectively. The distance value is 2.781671746 km, and (d) the green circle that buffers around the fourth serial shows the predicted distance value by the eNARX model. The predicted distance value from serial four is 0.2067022881 km.

(for Selangor area) and (b) recurrence of four serial criminal incidents (for Wilayah Persekutuan area) utilizing using Google Earth.

The first experiment showed that the prediction of the next crime location, time, and suspect was examined using two dynamic neural network methods. In this study, the NARX and BPTT methods were applied during the classification stage using MLBP, which was verified through MATLAB in predicting and modeling the crime time for serial cases. The two models of NARX and BPTT were examined in finding a more accurate method for the prediction of the next crime location, time, and suspect. Next, the selected method in the first part of the study was chosen to be enhanced using the objective functions of the RBF to achieve higher accuracy.

To achieve the goal of this study, repetitions three and four, which were produced by combining the “Suspect & Capture” and “Crime Plotting” datasets, were used to predict the distance for the commercial serial criminal. QGIS was applied to illustrate the difference between the predicted values of the next crime distance with the actual values of this study for the enhanced NARX, which provides a better understanding of the entire concept.

The observation of the outcome of this study demonstrates that the NARX model produces higher accuracy for crime prediction compared to BPTT. Thus, the NARX model was selected to be enhanced using SiRBF and was then compared with the original NARX model. The comparison results between the NARX and eNARX models indicate that the eNARX model shows higher accuracy based on the means and standard deviations.

Moreover, eNARX is a more appropriate method for predicting the next crime location, time, and suspect, due to its lower mean square error, the predicted value of the next crime location, time, and suspect as closer to the actual value. The average accuracy of the crime spatial datasets for the reliability of the eNARX, NARX and BPTT model is approximately 80.76%, 76.03%, and 37.82%, respectively, with a significant p-value of 0.012537 ($p < 0.05$), and the F value is 7.407918, which is higher than the F critical value of 4.256495.

D. EXPERIMENTAL RESULTS II

In this section, the three benchmark datasets of Dow Jones Index obtained from UCI machine repository, Monthly River Flow in the unit of cubic meters per second (cms), and Daily Temperature ($^{\circ}\text{C}$) get from the time series data library of data market (Table 5). These datasets present another justification that the intended model is also suitable for forecasting the next percent change of the price on the next week, the next cms of river flow for each month, and the next $^{\circ}\text{C}$ in temperature, respectively.

Table 5 also tabulates the total number of instances with 750, 588, and 1096 for the Dow Jones Index, Monthly River Flow in cubic meters per second (cms), and Daily Temperature ($^{\circ}\text{C}$) datasets in order. Those datasets contain about 14, 2, and 2 attributes, respectively. Dow Jones Index dataset possesses attributes, namely quarter, stock, date, and percent

TABLE 5. Time series benchmark datasets descriptions.

Name of Datasets	Dow Jones Index	Monthly River Flow (cms)	Daily Temperature (°C)
Dataset	Time series	Time series	Time series
Characteristics	Real	Real	Real
Attribute	Integer		
Instances value	750	588	1095
Attributes number	14	2	2
Area	Business	Hydrology	Meteorology

change of next week's price, whereas Monthly River Flow in cubic meters per second (cms) dataset dictates attributes like date and cubic meters per second. The third dataset of Daily Temperature (°C) keeps the attributes of date, degree as well as centigrade. Then these benchmark datasets were utilized to forecast the upcoming percent change of preceding week's price, the upcoming value of cubic meters per second, and the following degree centigrade temperature subsequently.

1) EXPERIMENTAL SETUP

The existing NARX and BPTT dynamic neural network methods were implemented in MATLAB to characterize and verify the MLBP classification stage. The examination was engaged in predicting the next percent change of the price on the next week, the next (cms) of river flow for each month and the next °C in temperature, for the Dow Jones Index, Monthly River Flow in cubic meters per second (cms) and Daily Temperature (°C) datasets respectively.

The Dow Jones Index dataset has inspected with numerous secreted neurons of 70 until 75. Whereas, the Monthly River Flow in cubic meters per second (cms) and the Daily Temperature (°C) datasets have been examined by numerous hidden neurons of 65 until 70 with a default iteration value of 1000 subsequently discarding the outliers of the data. The divisions for training, testing, and validation have been set at 70%, 15%, and 15%, respectively.

The Dow Jones Index dataset was examined when hidden neuron numbers varied from 70 to 75. Whereas, the Monthly River Flow in cubic meters per second (cms) datasets and Daily Temperature (°C) attributes have been examined when hidden neurons numbers varied from 65 to 70, with a default iteration value of 1000 subsequently discarding the outliers in the data. The divisions for training, testing, and validation have been set at 70%, 15%, and 15% respectively.

For the BPTT network, we built two feed-forward networks in the hidden layer, which consisted of Tansig, and one output neuron, which included a linear transfer function in the output layer for each network. To develop the NARX and eNARX networks, we constructed only two feed-forward networks in the hidden layer, which consisted of Tansig and SiRBF subsequently, and one output neuron that contained a

linear transfer function in the output layer for each network. To effectively train these networks, the standard LM algorithm applied with a learning rate of 0.001 and a regularization parameter of 1.00e-05, up to 1000 epochs. The quartile method has also employed to delete the outliers from every dataset as mentioned earlier of Dow Jones Index, Monthly River Flow in cubic meters per second (cms), and Daily Temperature in degree centigrade (°C).

Similarly, the existing BPTT, NARX, and proposed eNARX models have tested for their efficacy on those filtered. Three characteristics have utilized in the training set, and one character has used in the target set for the Dow Jones Index dataset. Whereas, for the Monthly River Flow in cubic meters per second (cms) and daily temperature in °C datasets, one characteristic was employed in the training set and other characteristics were used in the target set.

2) EXPERIMENT AND ANALYSIS FOR BENCHMARK DATASETS

The evaluation of the BPTT, NARX, and eNARX models indicated that the eNARX model offers higher accuracy compared to the other two models as per their mean and standard deviations. For the performance estimation of eNARX, NARX, and BPTT models, these three benchmark datasets have been applied in the training, validation, and testing stages. We can observe their relative accuracy with their means and standard deviations of these three datasets after applying BPTT, NARX, and the eNARX models as tabulated in Table 6. Fig.14 displays the precision values of the BPPT, NARX, and eNARX models.

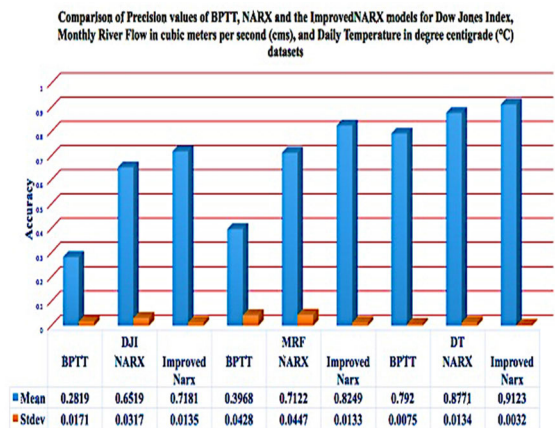


FIGURE 14. The precision score of the BPTT, NARX, and the eNARX models after predicting on Dow Jones Index, Monthly River Flow in cubic meters per second (cms), and Daily Temperature in degree centigrade (°C) datasets respectively.

For comparison purposes, the eNARX model has presented a higher accuracy compared to others in terms of their means and standard deviations. As a brief, existing NARX and BPTT models have been used and verified through MATLAB for prediction purposes.

To determine the prediction score of the upcoming percent change of the following week's price, upcoming cubic per second of river flow per month, and upcoming temperature (°C), NARX and BPTT models were inspected and analyzed. Next, this study utilizes an activation function mechanism, namely RBF, to obtain higher accuracy. In achieving the goal of this research, three benchmark datasets have been used. The primary results of the study indicate that the NARX network has shown more improved results compared to the BPTT model. Therefore, the NARX model has been selected to be enhanced through the fusion of (Tansig) with (RBF), which was then compared with the NARX model that used only Tansig.

The comparison of the results between these two mentioned models indicates that the eNARX model presents a higher accuracy compared to the NARX model, concerning their means and standard deviations. The average accuracy for all of the three models with the three benchmarks is approximately 82.15% for the enhanced NARX, 74.21% for NARX and 49.71% for BPTT. Here, a considerable p-value of $3.75E-25$ ($p < 0.05$) and an F value of 856.7246, which is greater than the F critical value of 3.354131, have been found. Furthermore, because of the lower mean square error that provides a closer prediction value to the actual value, eNARX model is the most appropriate model for the forecast of the next percent change of the next week's price, the next cubic meters per second of river flow per month and the next degree centigrade temperature.

V. RESULTS AND DISCUSSIONS

Due to the existence of time delay, the TDNN classifier such as NARX, and BPTT unable to identify the dynamic system by using single-activation functions. Therefore, utilizing the appropriate activation function can cause to tackle this issue [23]. Hence, this research suggests the fusion of two activation functions namely hyperbolic tangent (Tansig) and Radial Basis Function (RBF) in the same hidden layer.

In this part, the efficiency of the proposed eNARX in comparison to the NARX and the BPTT models is evaluated based on applying these models on two crime spatial datasets (repetition three and four) and three benchmark datasets namely, Dow Jones Index, Monthly River Flow in cubic meters per second (cms), and Daily Temperature in degree centigrade (°C).

To evaluate these techniques different hidden numbers value from 10 up to 100 for each of these datasets were tested. This evaluation demonstrated that the best hidden layer value for the two crime spatial datasets of repetition three and four is between 65 to 70. While, the best hidden neurons for Dow Jones Index is between 70 until 75, and 65 until 70 is the most proper hidden layer value for Monthly River Flow in cubic meters per second (cms), and Daily Temperature in degree centigrade (°C). A delay value for all datasets set to four, and all datasets allocated into three segments for training, validation, and testing at 70%, 15%, and 15% respectively [36]. Moreover, the classification of these models is checked through MATLAB.

TABLE 6. Evaluation of performance precision for Dow Jones index, monthly river flow (CMS), and daily temperature (°C) datasets for BPTT, NARX And eNARX models.

Dataset		BPTT Model				NARX Model				eNARX Model			
Dow Jones Index	Number of Hidden Neurons	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²
	70	0.2552	0.9696	0.9847	0.0651	0.6008	0.7778	0.8819	0.3610	0.7413	0.4687	0.6846	0.5495
	71	0.2784	0.9227	0.9606	0.0775	0.6323	0.7565	0.8698	0.3998	0.7062	0.5412	0.7357	0.4987
	72	0.3041	0.9121	0.955	0.0925	0.6924	0.6544	0.8089	0.4794	0.7128	0.5102	0.7143	0.5081
	73	0.2758	0.9324	0.9656	0.0761	0.6671	0.6682	0.8174	0.4450	0.7095	0.5291	0.7274	0.5034
	74	0.2814	0.9214	0.9599	0.0792	0.6649	0.6701	0.8186	0.4421	0.7116	0.5152	0.7178	0.5064
	75	0.2962	0.9176	0.9579	0.0877	0.6540	0.6864	0.8285	0.4277	0.7273	0.5050	0.7106	0.5290
	Mean	0.2819				0.6519				0.7181			
	Stdev	0.0171				0.0317				0.0135			
Monthly River Flow in cubic meters per second (cms)	Number of Hidden Neurons	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²
	65	0.4397	0.8076	0.8987	0.1933	0.7755	0.4941	0.7029	0.6014	0.8115	0.3490	0.5908	0.6585
	66	0.4317	0.8146	0.9025	0.1864	0.7342	0.5662	0.7525	0.5390	0.8461	0.2862	0.5349	0.7159
	67	0.3904	0.8535	0.9238	0.1524	0.6821	0.6759	0.8221	0.4653	0.8345	0.3306	0.5750	0.6964
	68	0.3232	0.9374	0.9682	0.1045	0.7365	0.5412	0.7357	0.5424	0.8235	0.3225	0.5766	0.6782
	69	0.3798	0.8801	0.9381	0.1442	0.6518	0.6582	0.8113	0.4248	0.8207	0.3346	0.5785	0.6735
	70	0.4159	0.8316	0.9119	0.1730	0.6929	0.6675	0.8170	0.4801	0.8128	0.3440	0.5865	0.6606
	Mean	0.3968				0.7122				0.8249			
	Stdev	0.0428				0.0447				0.0133			
	Daily Temperature in Degree Centigrade (°C)	Number of Hidden Neurons	R	MSE	RMSE	R ²	R	MSE	RMSE	R ²	R	MSE	RMSE
65		0.8015	0.3573	0.5978	0.6424	0.877	0.2361	0.486	0.7691	0.9087	0.1768	0.4205	0.8257
66		0.7907	0.3747	0.6121	0.6252	0.8860	0.2210	0.4802	0.7805	0.9090	0.1747	0.4180	0.8263
67		0.7992	0.3647	0.6039	0.6387	0.8656	0.2619	0.5118	0.7493	0.9114	0.1707	0.4131	0.8307
68		0.7810	0.3908	0.6252	0.6100	0.8566	0.2888	0.5374	0.7338	0.9163	0.1614	0.4018	0.8396
69		0.7879	0.3789	0.6156	0.6208	0.8898	0.2144	0.4630	0.7917	0.9157	0.1619	0.4024	0.8385
70		0.7915	0.3733	0.6110	0.6265	0.8874	0.2148	0.4635	0.7875	0.9124	0.1685	0.4105	0.8325
Mean		0.7920				0.8771				0.9123			

The evaluation of the proposed eNARX with NARX and BPTT showed that the proposed eNARX outperforms other techniques in terms of higher accuracy.

The significant t-Test utilized to show the reliability of proposed eNARX in comparison to the NARX and the BPTT model. The average precisions for the reliabilities of the commercial crime series for the eNARX, NARX, and BPTT models are approximately 80.76%, 76.03%, and 37.82%, respectively, with a significant p-value of 0.0125 ($p < 0.05$). In contrast, the three benchmark datasets in the fields of Business, Hydrology, and Meteorology achieved approximately 82.15%, 74.21%, and 49.71%, respectively, with a significant p-value of $3.75E-25$ ($p < 0.05$).

VI. CONCLUSION AND FUTURE WORK

In this work, we studied the time delay associated with time delay neural network (TDNN), which caused the uncertainty and poor performance of TDNN classifiers. The BPTT and NARX models are considered a well-known TDNN classifier system. As the activation function has a major impact on the TDNN classifier's performance, the creation of an appropriate activation function is useful for the identification of the dynamic systems.

Hence, this research proposed a novel technique of eNARX classifier based on the fusion of two activation functions namely, hyperbolic tangent (Tansig) and Radial Basis Function (RBF) in the same hidden layer (SiRBF), and design the obtained forecast using eNARX model for the commercial serial crime spatiotemporal. These functions are advantageous for being nonlinear and having robust against input noises. Therefore, the proposed method can significantly affect the TDNN error minimization. It also produces better forecasting for crime spatiotemporal with an adaptation of dynamic system.

The proposed eNARX model based on the fusion of two activation functions (SiRFB) is beneficial to tackle the issue of uncertainty and poor performance of TDNN classifiers which occurred due to time delay. The experimental results of this study showed that the proposed eNARX model outperforms other techniques of BPTT and NARX in terms of producing higher precision for prediction of next serial crime. However, the proposed eNARX has higher computational complexity compare to BPTT model as it has complex structure.

This work was carried out with activation functions in the same hidden layer for prediction of next serial crime in conventional neural network, hence future predictions can be considered based on the use of activation functions in different hidden layers in deep networks.

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